

# Federated Commerce on nopCommerce

## Scenario A · Architectural Evolution After Acquisitions

Software Architecture · Group Assignment 02 · Part 1

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# The Architectural Tension

*Northstar Living Group: a retail holding that grew by acquisition.  
Two specialist BUs need a shared platform without losing local autonomy.*

## PULL • CENTRALISATION

- ✓ Single sign-on across the group
- ✓ One platform
- ✓ Group-level customer view and analytics

## PULL • ISOLATION

- ✓ BU-specific pricing and catalog rules
- ✓ Local ERP, stock, order workflows
- ✓ Independent deployability, no failure crossover

**No neutral position.** Every share-vs-isolate choice has consequences for coupling, failure propagation, and BU autonomy.

# Why It Matters: Business Reality

|                                                                                                                                                               |                                                                                                                                                                      |                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>COST OF GETTING IT WRONG</div> <div><b>Force one rigid model</b><br/>Lose brand differentiation and the customer loyalty each acquisition brought.</div> | <div>COST OF GETTING IT WRONG</div> <div><b>Allow drift</b><br/>Recurring outages and slow delivery erode the cost savings consolidation was meant to produce.</div> | <div>WHAT LEADERSHIP WANTS</div> <div><b>Shared foundation, local autonomy</b><br/>Lower platform cost. Group insights. Preserve what makes each brand work.</div> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|

The architecture IS the business decision: what to share, what to isolate, where the seams sit.

# Why nopCommerce Makes This Interesting

STORE-AWARE. BUT NOT STORE-ENFORCED.

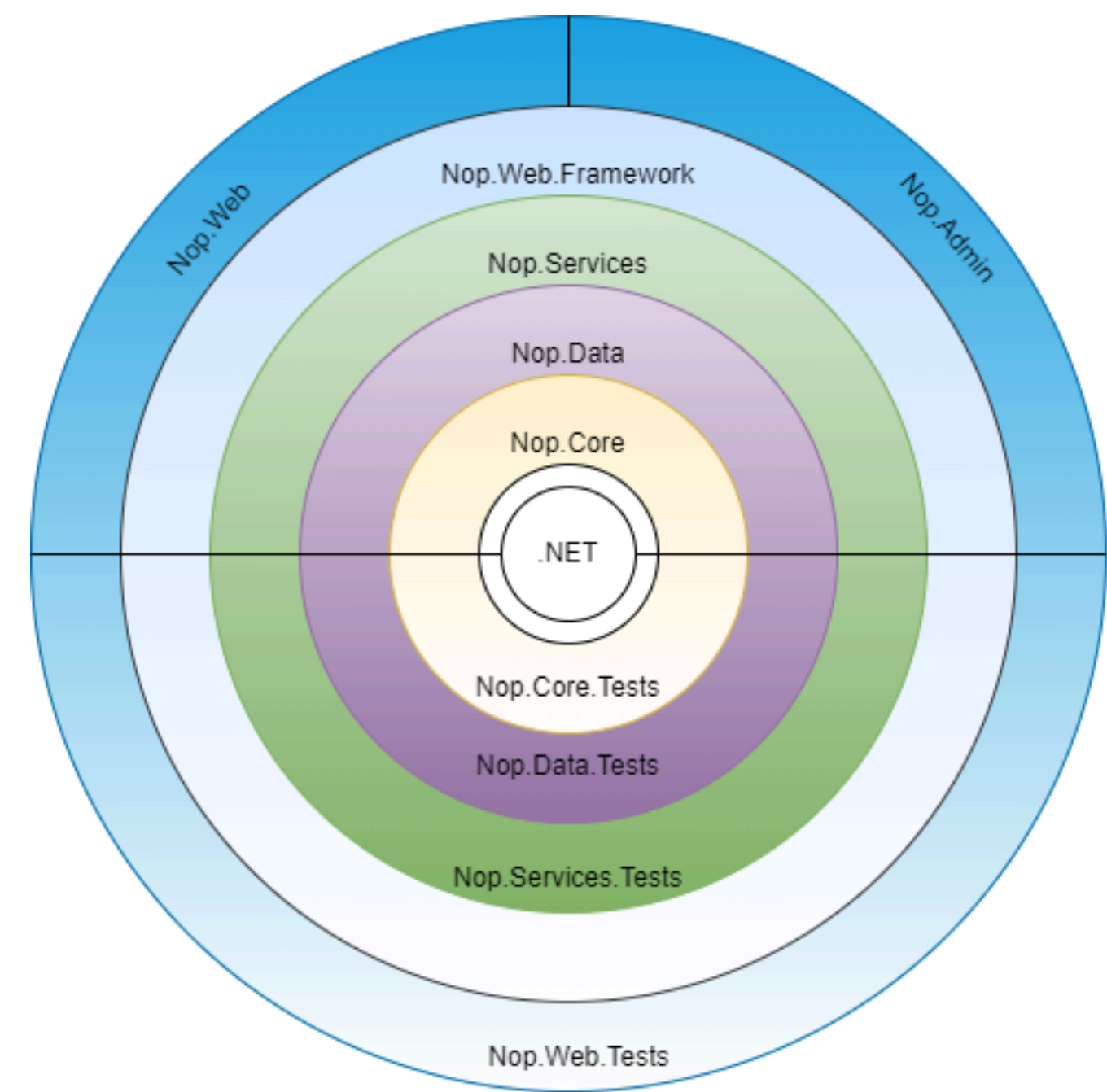
## What it gives us

- Multi-store concept
- Per-store settings with global fallback
- StoreMapping bridge
- ExternalAuthenticationMethod
- IEventPublisher with auto entity events.

## What it does not

- Customer identity has no store dimension
- Roles have no store scope
- Inventory is warehouse-global
- Catalog isolation is opt-in
- Events are in-process, sequential, non-durable.

# Current Architecture · Onion Modular Monolith



## SINGLE DEPLOYABLE

One ASP.NET Core process, one relational DB. Plugins share the DI container. No hard boundaries between layers.

A memory leak, slow plugin, or runaway query in **BU1** degrades **BU2** in the same instance.

Each BU must run as its own nopCommerce process. Process-level isolation is the only honest path.



# Multi-Store Today: Opt-In Everywhere

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## BUILT IN

- ✓ Per-store URLs, themes, settings
- ✓ Per-store settings with global fallback
- ✓ Orders stamped with StoreId
- ✓ TierPrice can be per-store
- ✓ StoreMapping bridge for catalog

## HIDDEN ASSUMPTIONS

- ✗ **Without explicit mapping**, a product appears in every store
- ✗ **IgnoreStoreLimitations** is a global bypass switch
- ✗ **No store group**, hierarchy, or parent-child concept
- ✗ **Settings change globally** with no approval gate

# Pressure Points · Where the Code Fights Federation

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## P1 · PRICING

### **Single global product price**

BUs cannot own pricing without parallel TierPrice per product per BU.

## P2 · IDENTITY

### **Customer identity is fully global**

No StoreId on Customer or CustomerRole. A BU1 customer has implicit access to BU2.

## P3 · INVENTORY

### **Warehouse-centric, not BU-aware**

All stores draw from the same pool. A stock-out in BU1 is a stock-out everywhere.

## P4 · EVENTS

### **IEventPublisher is in-process**

Sequential, non-durable. Cannot cross processes. Federation needs a broker.

## P6 · BYPASS

### **IgnoreStoreLimitations**

A single admin toggle silently disables every catalog isolation rule across the group.

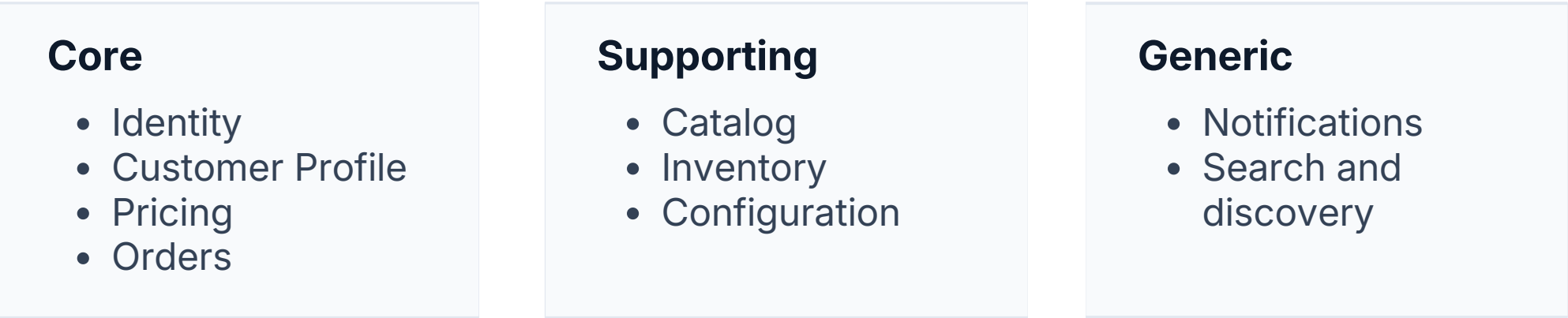
## P7 · ACL

### **Role-scoped only, not store-scoped**

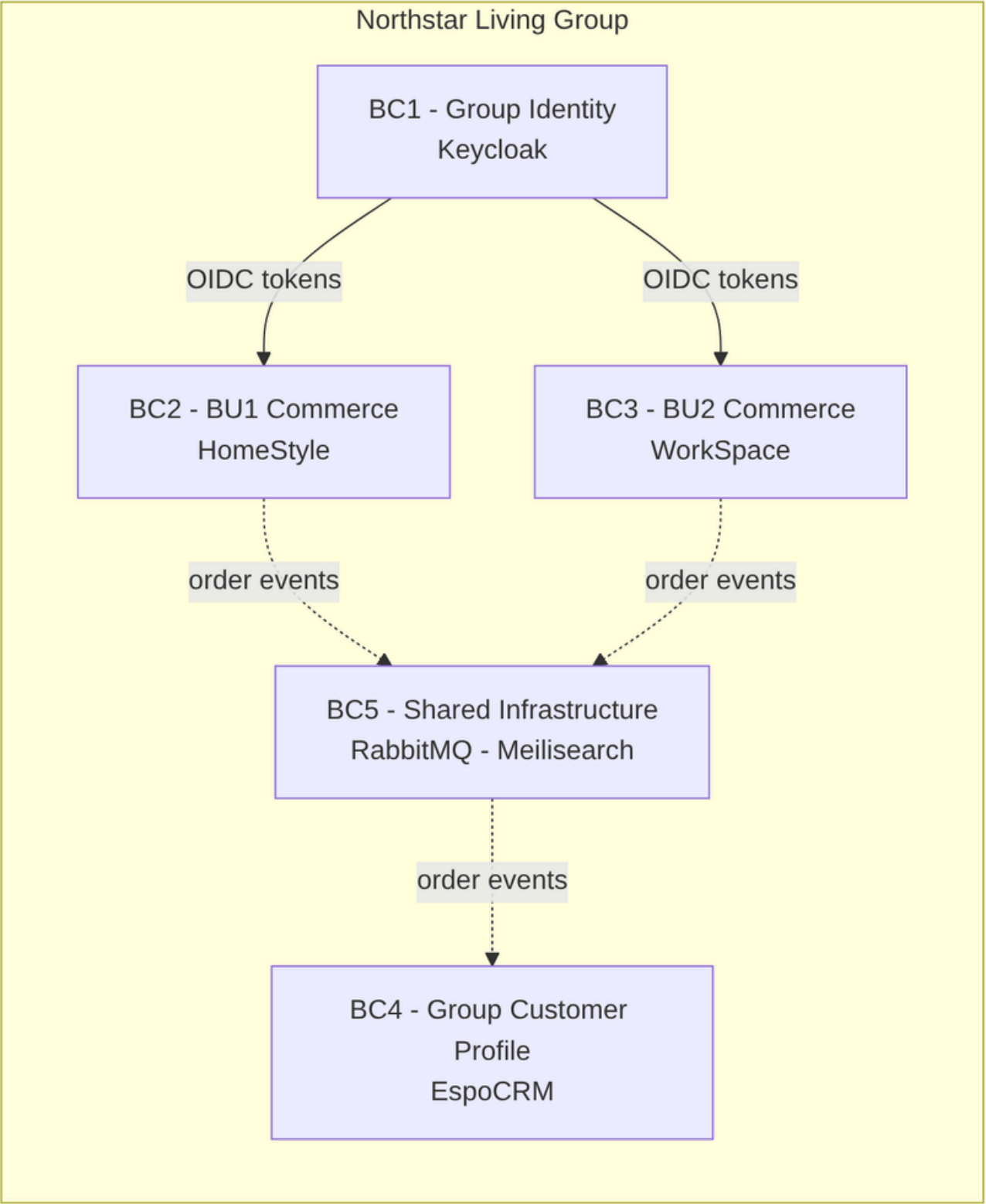
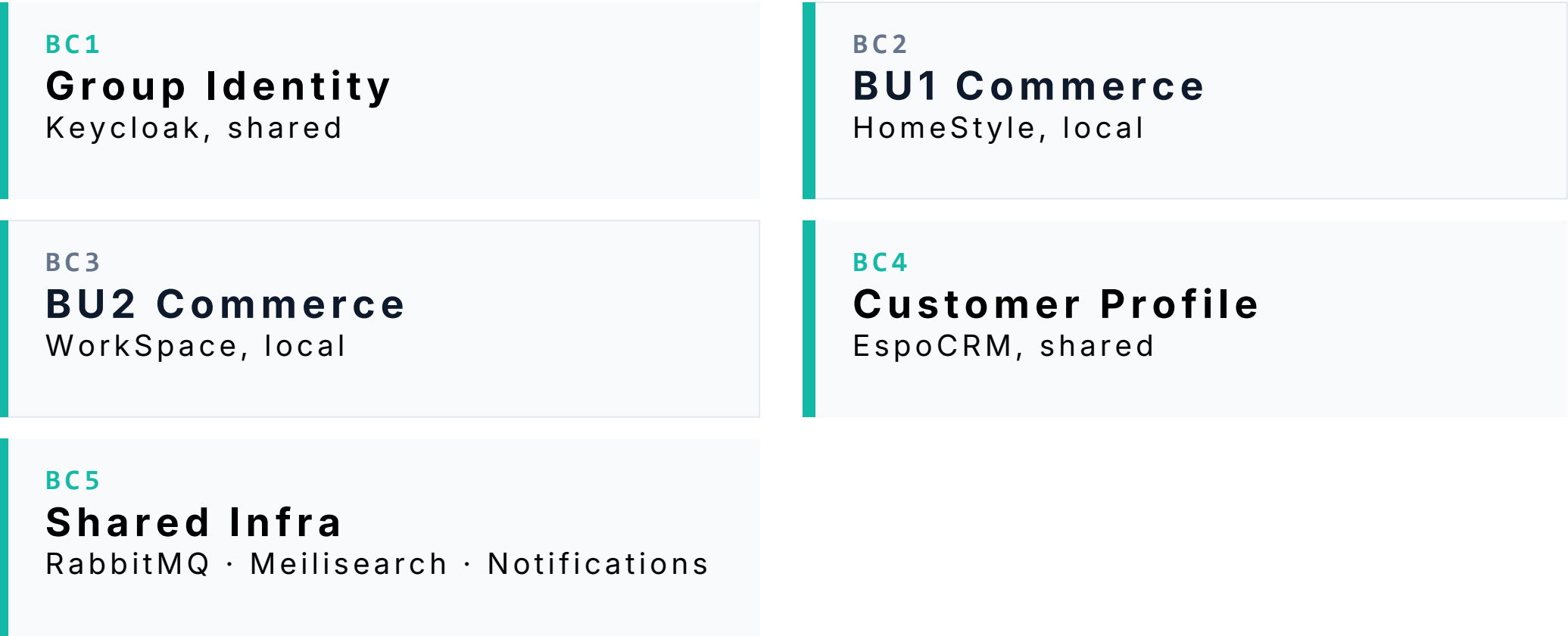
A product visible to a wholesale role is visible in every store where that role exists.

# Domain · Subdomains and Bounded Contexts

## DDD SUBDOMAINS



## FIVE BOUNDED CONTEXTS





# Ownership · One Rule

Anything that can be owned by a single BU IS owned by that BU. Anything that requires a cross-BU view sits in a group context.

| RESPONSIBILITY                        | OWNER           | MODE                                 |
|---------------------------------------|-----------------|--------------------------------------|
| Customer authentication, group claims | BC1 Keycloak    | Synchronous · OIDC                   |
| Catalog, pricing, stock, orders       | BC2 / BC3       | Local, autonomous                    |
| Cross-BU customer profile             | BC4 EspoCRM     | Async read model · never writes back |
| Federated search index                | BC5 Meilisearch | Async re-index from broker           |
| Event transport                       | BC5 RabbitMQ    | Durable · BU-tagged                  |

**BUs never call each other.** All cross-BU coordination is mediated by BC5.

# Quality Attribute Scenarios · Five Drivers

QA1 · AVAILABILITY · ITER 1, 4 · ADR-001, ADR-004

**Fault Isolation**  
BU2 fails (process, DB, or ERP): BU1 untouched (p95 ≤ 300 ms). BU2 stays open with cached stock within 5 s.

QA2 · MODIFIABILITY · ITER 1 · ADR-001

**BU Pricing Autonomy**  
BU1 reprices 200 products: visible in ≤ 60 s. BU2 unchanged. No shared setting touched.

QA3 · SECURITY · ITER 2 · ADR-002

**Federated Identity**  
SSO across BUs. Session transition under 2 s. Zero role bleed-through.

QA4 · RELIABILITY · ITER 5 · ADR-005

**Search Degradation**  
Meilisearch down: fallback active in 3 s, results in 5 s, resync in 5 min, no permanent loss.

QA5 · CONSISTENCY · ITER 3 · ADR-003

**Cross-BU Order Visibility**  
Order on BU2 reaches CRM in ≤ 30 s. Survives 24h consumer outage. No duplicates after replay.

# Framework · Attribute-Driven Design

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WHY   ADD

1

## **QA-driven by design**

Each scenario in QA1-QA5 maps to ADD iterations.

2

## **Brownfield ready**

Asks what is already there and what must change. No blank-slate assumption.

3

## **Traceability built in**

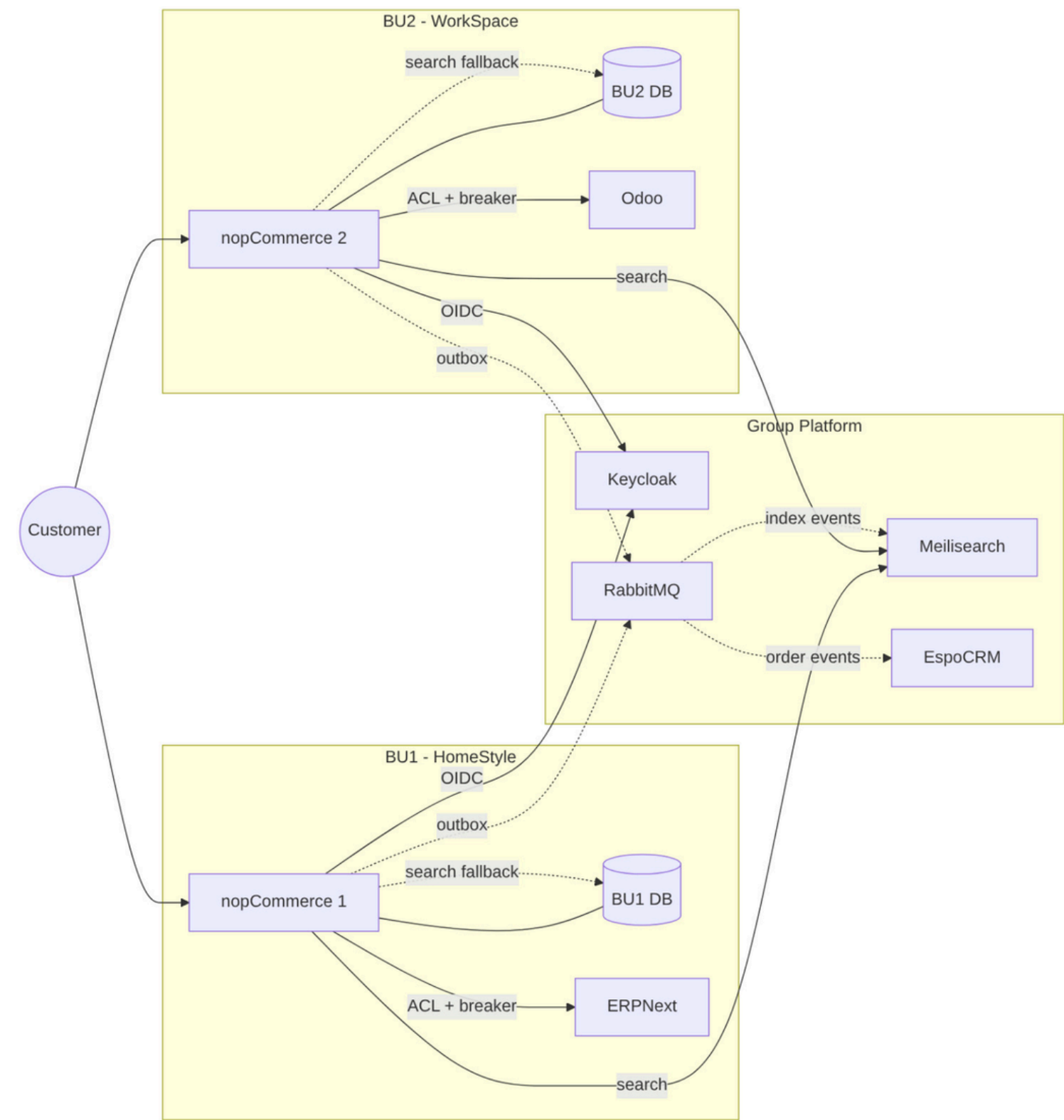
Every ADR references QA + pressure point + BC + iteration.

4

## **Bounded by drivers**

Stop when QAs are satisfied, not when ideas run out.

# Target Architecture · Component View



Solid lines: synchronous. Dotted lines: async via broker. **Eight independently deployable subsystems · no shared DB across extracted boundaries.**

# Cross-Cutting Concerns

ADR-001 · 003 · 005

## BU scoping

Every row, event payload, and search document carries a BU + store identifier.

ADR-002 · 001

## Identity & authz

Single Keycloak issuer. Group claims in JWT. Per-BU role mapping is config-as-code, BU-prefixed.

ITER 2-5

## Observability

Logs include BU, correlation id, subject. Metrics on breaker state, outbox lag, fallback rate, consumer lag.

ADR-004 · 005

## Reliability primitives

One Polly policy reused across ERP, broker publish, and search read.

QA5 · ADR-003

## Consistency model

Strong inside each BU. Eventual between BUs and to group services. Slips are visible, never silent.

ADR-004 · 005

## Failure-mode UX

"Stock to be confirmed", "limited results". No silent fallbacks.



# Per-BU Process and Database Isolation

QA1 · QA2   P1 · P6 · P7   **BC2 · BC3**   ITER 1

## DECISION

Each BU runs as its own ASP.NET Core 9 process with its own PostgreSQL database. No shared process state, DI container, GC, connection pool, or relational DB. The multi-store feature still works inside each instance but stops being load-bearing for cross-BU isolation.

## REJECTED ALTERNATIVES

✗ **One instance + mandatory StoreMapping.** Was the prior proposal. Satisfies QA2 but not QA1: a single instance is one failure domain.

# Federate Identity via Keycloak

QA3

P2

BC1

ITER 2

## DECISION

Adopt Keycloak as group IdP. Each nopCommerce BU is an OIDC relying party via a custom plugin implementing `IEternalAuthenticationMethod`. Authorization Code + PKCE. 5 min access token, 30 min refresh, sliding session. Group claims travel in JWT; BU-local roles mapped from claims on first login per BU.

## REJECTED ALTERNATIVES

✗ **Replicate customers via events.** Eventual consistency on identity becomes a correctness bug, not just latency.

✗ **Hosted IdP (Auth0, Cognito, Azure AD B2C).** Architecturally identical. Vendor lock-in unwarranted.

# Bridge IEventPublisher to RabbitMQ via Outbox

QA5

P4

BC5

ITER 3

## DECISION

Bridge IEventPublisher to RabbitMQ through a transactional outbox. Order row + outbox row written in the same DB transaction. Relay reads unpublished rows in order and marks them published. Topic exchange per concern. Routing keys carry the BU. Durable queues, manual ack, dead-letter. Idempotency on (aggregate\_id, version).

## REJECTED ALTERNATIVES

- ✗ **Synchronous HTTP from BU to CRM.** Couples order placement to CRM availability.
- ✗ **Kafka.** Streaming and replay are not the shape of "process each order once with retries" at our volume.



# BU-Local ERP · ACL · Circuit Breaker



- QA1
- P3
- BC2 · BC3
- ITER 4

## DECISION

Each BU runs its own ERP (ERPNext for BU1, Odoo for BU2). Each nopCommerce instance integrates through a dedicated Anti-Corruption Layer with a Polly circuit breaker. Open after consecutive failures or sustained latency. Half-open probe recovers automatically. When open: cached stock snapshot (TTL 5 min) + "stock to be confirmed at fulfilment" UX.

## REJECTED ALTERNATIVES

- ✗ **Single shared ERP.** Reproduces the monolith problem one layer down.
- ✗ **Plain timeout instead of breaker.** Still hits the broken ERP on every request.

# Federated Meilisearch · DB Fallback Strategy

QA4

P4 · P1

BC5

ITER 5

## DECISION

Meilisearch as federated search index, populated asynchronously via ADR-003's broker. Storefront search runs through a strategy pattern with circuit breaker: Meilisearch primary, BU-local DB fallback. Documents tagged with originating BU. 1.5 s timeout. Half-open probe at 30 s. Recovery drains backlog from RabbitMQ.

## REJECTED ALTERNATIVES

✗ **Elasticsearch.** Same shape, much heavier ops cost (JVM, multi-node baseline, more knobs).

# Risks

R2 · P=2 · I=3 · SCORE 6 · ADR-003

## Outbox / idempotency bug

Mitigation: Transactional outbox. Idempotent consumer on (aggregate\_id, version). Test simulating relay crash mid-publish.

R3 · P=3 · I=2 · SCORE 6 · ADR-001

## BU deployment drift over time

Mitigation: Shared deployment templates and CI per BU. Quarterly version-sync calendar. Integration test asserts both BUs run the same nopCommerce version.

R9 · P=3 · I=2 · SCORE 6 · ADR-001, ADR-005

## Schema drift between BUs breaks federated search

Mitigation: Indexer ACL normalises BU schemas to one search-document shape. Integration test per BU mapping.

# Feasibility Spike · ADR-002 Proven End-to-End

## WHY THIS DECISION

The OIDC plugin is the most consequential new external component, and we had not proven nopCommerce would accept it through the existing `IExternalAuthenticationMethod` seam. Federated identity is the riskiest decision because it is the one piece of cross-BU shared state that must stay in lock-step.

## ACCEPTANCE CRITERIA

- AC1 - Keycloak login on BU1 redirects and authenticates successfully.
- AC2 - Navigating to BU2 in the same session requires no login credentials and recognises the user with the correct role.
- AC3 - Each BU has exactly one Customer row sharing the same Keycloak sub, with independently mapped roles.

AC1 PASS

AC2 PASS

AC3 PASS

# Evolution Roadmap · Five Iterations

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1

## **Per-BU process and database isolation** [QA1](#) · [QA2](#) · [ADR-001](#)

Split the single deployable into one ASP.NET Core process + one PostgreSQL DB per BU.

2

## **Federated identity via Keycloak** [QA3](#) · [ADR-002](#)

Keycloak HA pair. Each instance becomes an OIDC relying party. Group claims map to BU-prefixed local roles on first login.

3

## **Cross-BU async coordination** [QA5](#) · [ADR-003](#)

RabbitMQ + outbox + relay inside each BU. EspoCRM consumes order events idempotently. In-process publisher keeps working for in-process consumers.

4

## **Per-BU ERP behind ACL and breaker** [QA1](#) · [ADR-004](#)

Local ERP per BU. ACL + Polly breaker. Cached stock snapshot serves the open-circuit fallback. Second layer of fault isolation.

5

## **Federated search with fallback** [QA4](#) · [ADR-005](#)

Meilisearch populated via the broker stream from step 3. Strategy + breaker falls back to BU-local DB search. DB path stays permanently.



**Questions?**